Solve with us 11 - Not Graded

This assignment will not be graded and is only for practice.
Higher order partial derivatives:Higher order partial derivatives:

- Let ff be a scalar valued function defined on a domain DD in R^2 R2. Then the second order partial derivatives are the partial derivatives of the partial derivatives.

Notations:Notations:

1. The partial derivative of $\frac{\partial f}{\partial x}$∂x∂f with respect to xx is denoted by $f_{xx} = \frac{\partial^2 f}{\partial x^2}$fxx = ∂x2∂2f.
2. The partial derivative of $\frac{\partial f}{\partial y}$∂y∂f with respect to xx is denoted by $f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$.fyx = ∂x∂y∂2f.
3. The partial derivative of $\frac{\partial f}{\partial x}$∂x∂f with respect to yy is denoted by $f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$.fxy = ∂y∂x∂2f.
4. The partial derivative of $\frac{\partial f}{\partial y}$∂y∂f with respect to yy is denoted by $f_{yy} = \frac{\partial^2 f}{\partial y^2}$fyy = ∂y2∂2f.

- Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a function defined on a domain DD in R^n Rn. Then the second order partial derivatives of ff are defined analogously as the partial derivatives of the partial derivatives.

Notation:Notation: The partial derivative of $\frac{\partial f}{\partial x_i}$∂xi∂f with respect to x_j xj is denoted as $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$ fxixj = ∂xj∂xi∂2f.

Note:Note: In general, the mixed partial derivatives $f_{x_i x_j}$ fxixj and $f_{x_j x_i}$ fxjxi are not necessary equal.

Clairaut's_Theorem: Clairaut's Theorem :Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a scalar valued function defined on a domain DD containing a point $a\sim = (a_1, a_2, \ldots, a_n)$~a = (a1, a2, … , an) in R^n Rn. If both $f_{x_i x_j}$ fxixj and $f_{x_j x_i}$ fxjxi are defined and continuous in an open ball containing the point $a\sim = (a_1, a_2, \ldots, a_n)$~a = (a1, a2, … , an), then

$$f_{x_i x_j}(a\sim) = f_{x_j x_i}(a\sim).\text{fxixj}(\sim\text{a}) = \text{fxjxi}(\sim\text{a}).$$

$k$kth-order partial derivatives:th-order partial derivatives: Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a function defined on a domain DD in R^n Rn. Then the $k$kth-order partial derivatives of ff are defined analogously by taking successive partial derivatives for kk-times.

Notation:Notation: The notation for $k$kth-order partial derivative is

$$f_{x_{i_1} x_{i_2} \ldots x_{i_k}} = \frac{\partial^k f}{\partial x_{i_k} \ldots \partial x_{i_2} \partial x_{i_1}}\text{fxi1 xi2 …xik} = \partial\text{xik} \ldots \partial\text{xi2}\,\partial\text{xi1}\,\partial\text{kf}$$

Modified statement of Clairaut's Theorem: Modified statement of Clairaut's Theorem :Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a scalar valued function defined on a domain DD containing a point $a\sim = (a_1, a_2, \ldots, a_n)$~a = (a1, a2, … , an) in R^n Rn whose $k$kth-order partial derivatives are all continuous in an open ball containing the point $a\sim = (a_1, a_2, \ldots, a_n)$~a = (a1, a2, … , an). For any integers $i_1, i_2, \ldots, i_k$ i1, i2, … , ik between 11 and nn, if $j_1, j_2 \ldots j_k$ j1, j2 … jk is a reo of $i_1, i_2 \ldots i_k$ i1, i2 … ik, then

$f_{x_{i_1} x_{i_2} \ldots x_{i_k}}(a\sim) = f_{x_{j_1} x_{j_2} \ldots x_{j_k}}(a\sim)$.fxi1 xi2 …xik (∼a) = fxj1 xj2 …xjk (∼a).

A consequence of this theorem is that
we don't need to keep track of the order in which we take partial derivatives.
we don't need to keep track of the order in which we take partial derivatives.

Hessian matrix:Hessian matrix: Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a scalar valued function defined on a domain DD in R^n Rn. Then the Hessian matrix of ff is defined as

$$\mathrm{H}f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.\mathrm{Hf} = \begin{pmatrix} \partial x12\partial 2f\partial x2\partial x1\partial 2f & \vdots & \partial xn\partial x1\partial 2f\partial x1\partial x2\partial 2f\partial x22\partial 2f & \vdots & \partial xn\partial x2\partial 2f & \cdots & \cdots & \ddots & \cdots & \partial x1\partial xn\partial 2f\partial x2\partial xn\partial 2f & \vdots & \partial xn2\partial 2f \end{pmatrix}.$$

H$f(a_1, a_2 \ldots a_n)$Hf(a1, a2 … an) means that all of the entries of HfHf are evaluated at the point $(a_1, a_2 \ldots a_n)$(a1, a2 … an), which must be a point in the domain of ff.

1 point
Find $\frac{\partial^2 f}{\partial x \partial y}$∂x∂y∂2f of the function $f(x, y) = x^2 y^3 - xy + 6x + 8y + 1$.
f(x, y) = x2y3 − xy + 6x + 8y + 1.
○
$2xy^3 - y + 6$2xy3 − y + 6
○
$3x^2 y - x + 6$3x2y − x + 6
○
$6xy^2$ 6xy2-1
○
00
No, the answer is incorrect.
Score: 0
Accepted Answers:
$6xy^2$ 6xy2-1
1 point
Find $\frac{\partial^2 f}{\partial x^2}$∂x2∂2f of the function $f(x, y) = x^5 + xy^7 + e^{xy}$ f(x, y) = x5 + xy7 + exy at point $(1, -1)$.
(1, −1).
○
$20 + e$20 + e.
○

$20 - e^{-1}$ 20 − e−1.

20 − e20 − e.

20 + e^{-1} 20 + e−1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

20 + e^{-1} 20 + e−1.

1 point

Choose the correct statements about $f(x, y, z) = -x^4 + 2x^2 y^2 z^2 - z^2 y^4 + 2y^2 - xz^4 + 2z^2$.
f(x, y, z) = −x4 + 2x2y2z2 − z2y4 + 2y2 − xz4 + 2z2.

- [ ] $\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$ ∂x∂y∂z∂3f = 16xyz
- [ ] $\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz - 8zy^3$ ∂z∂y∂x∂3f = 16xyz − 8zy3
- [ ] $\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$ ∂z∂y∂x∂3f = 16xyz
- [ ] $\frac{\partial^4 f}{\partial z^4} = -24x$ ∂z4∂4f = −24x
- [ ] $\frac{\partial^5 f}{\partial y^5} = -24$ ∂y5∂5f = −24
- [ ] $\frac{\partial^5 f}{\partial y^5} = -24z^2$ ∂y5∂5f = −24z2

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz$ ∂x∂y∂z∂3f = 16xyz

$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz$ ∂z∂y∂x∂3f = 16xyz

$\frac{\partial^4 f}{\partial z^4} = -24x$ ∂z4∂4f = −24x

1 point

The determinant of the Hessian matrix of $f(x, y) = x^2 + 3xy + 7y^2$ f(x, y) = x2 + 3xy + 7y2 at point (x, y)(x, y) is

2828

1919

00

−19−19

No, the answer is incorrect.

Score: 0

Accepted Answers:

1919

1 point

Let AA denote the Hessian matrix of the function $f(x, y, z) = 3x^2 + xz + 2zy - z^2$
f(x, y, z) = 3x2 + xz + 2zy − z2 at (x, y, z)(x, y, z). Which of the following statements is (are) true?

- [ ] $A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$

- [ ] $A = \begin{pmatrix} -6 & 0 & -1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\begin{pmatrix} -6 & 0 & -1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$

- [ ] Det (A) = 24(A) = 24
- [ ] Det (A) = -24(A) = −24

No, the answer is incorrect.
Score: 0
Accepted Answers:

$A = \begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$A = $\begin{pmatrix} 6 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & -2 \end{pmatrix}$

Det (A) = -24(A) = −24

1 point

Let AA denote the Hessian matrix of the function $f(x, y, z) = x^2 y - y^3 z^2$ f(x, y, z) = x2y − y3z2 at (1, 1, 1)(1, 1, 1). Which of the following statements is (are) true?

- [ ] Rank (A) = 3(A) = 3
- [ ] Rank (A) = 2(A) = 2
- [ ] Nullity (A) = 1(A) = 1
- [ ] Nullity (A) = 0(A) = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
Rank (A) = 3(A) = 3
Nullity (A) = 0(A) = 0

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two
The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two
Let $f(x, y)$f(x, y) be a function defined on a domain DD in R^2.R2. Suppose that $f(x, y)$f(x, y) and its first and second order partial derivatives are continuous in an open ball around $(a, b) \in D$(a, b) ∈ D and $f_x(a, b) = f_y(a, b) = 0$.fx(a, b) = fy(a, b) = 0. Then

1. ff has a local minimum at (a, b)(a, b) if $f_{xx}(a, b) > 0$fxx(a, b) > 0 and det(H$f(a, b)$) > 0 (Hf(a, b)) > 0.

2. ff has a local maximum at (a, b)(a, b) if $f_{xx}(a, b) < 0$fxx(a, b) < 0 and det $(\mathrm{H}f(a, b)) > 0$ (Hf(a, b)) > 0.
3. ff has a saddle point at (a, b)(a, b) if det$(\mathrm{H}f(a, b)) < 0$(Hf(a, b)) < 0.
4. The test is inconclusive at (a, b)(a, b) if det$(\mathrm{H}f(a, b)) = 0$(Hf(a, b)) = 0.

1 point

Consider the function $f(x, y) = x^3 + x^2 y - y$f(x, y) = x3 + x2y − y. Which of the following are correct?

- [ ] $(-1, \frac{3}{2})$(−1, 23) is a local minimum.
- [ ] $(-1, \frac{3}{2})$(−1, 23) is a saddle point.
- [ ] $(1, \frac{-3}{2})$(1, 2−3) is a local maximum.
- [ ] $(1, \frac{-3}{2})$(1, 2−3) is a saddle point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-1, \frac{3}{2})$(−1, 23) is a saddle point.

$(1, \frac{-3}{2})$(1, 2−3) is a saddle point.

1 point

Choose the correct statement about $f(x, y) = x^2 - y^3 - x^2 y + y$f(x, y) = x2 − y3 − x2y + y.

- [ ] $(0, \frac{1}{\sqrt{3}})$(0, 3 √ 1) is a saddle point.
- [ ] $(0, \frac{-1}{\sqrt{3}})$(0, 3 √ −1) is a local minimum.
- [ ] $(0, \frac{-1}{\sqrt{3}})$(0, 3 √ −1) is a local maximum.
- [ ] Hessian test in inconclusive at the point $(0, \frac{1}{\sqrt{3}})$(0, 3 √ 1).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, \frac{1}{\sqrt{3}})$(0, 3 √ 1) is a saddle point.

$(0, \frac{-1}{\sqrt{3}})$(0, 3 √ −1) is a local minimum.

1 point

Choose the correct statement about $f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x$.
f(x, y) = 2x3 + 6xy2 − 3y3 − 150x.

☐
(5, 0)(5, 0) is a local minimum.

☐
(− 3, − 4)(−3, −4) is a saddle point.

☐
(− 5, 0)(−5, 0) is a local minimum.

☐
(− 5, 0)(−5, 0) is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(5, 0)(5, 0) is a local minimum.
(− 3, − 4)(−3, −4) is a saddle point.

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of thr
The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of thr
Let $f(x, y, z)$f(x, y, z) be a function defined on a domain DD in R^3.R3. Suppose that $f(x, y, z)$
f(x, y, z) and its first and second order partial derivatives are continuous in an open ball around
$(a, b, c) \in D$(a, b, c) ∈ D and $f_x(a, b, c) = f_y(a, b, c) = f_z(a, b, c) = 0$.fx(a, b, c) = fy
(a, b, c) = fz(a, b, c) = 0. Then

1. ff has a local minimum at (a, b, c)(a, b, c) if $f_{xx}(a, b, c) > 0$fxx(a, b, c) > 0,
   $(f_{xx} f_{yy} - f_{xy}^2)(a, b, c) > 0$(fxxfyy − fxy2)(a, b, c) > 0, and det(H$f(a, b, c)$) > 0(Hf(a, b, c)) > 0.
2. ff has a local maximum at (a, b, c)(a, b, c) if $f_{xx}(a, b, c) < 0$fxx(a, b, c) < 0,
   $(f_{xx} f_{yy} - f_{xy}^2)(a, b, c) > 0$(fxxfyy − fxy2)(a, b, c) > 0, and det(H$f(a, b, c)$) < 0(Hf(a, b, c)) < 0.
3. ff has a saddle point at (a, b, c)(a, b, c) if det(H$f(a, b, c)$) ≠ 0(Hf(a, b, c)) = 0 and cases 11 and 22 do not occur.
4. The test is inconclusive at (a, b, c)(a, b, c) if det(H$f(a, b, c)$) = 0(Hf(a, b, c)) = 0.

1 point
Consider the function $f(x, y, z) = x^3 + xz^2 - 3x^2 + y^2 + 2z^2$
f(x, y, z) = x3 + xz2 − 3x2 + y2 + 2z2. Which of the following options are correct?

○
(0, 0, 0)(0, 0, 0) is a local maximum.

○
(0, 0, 0)(0, 0, 0) is a saddle point.

○
Hessian test is inconclusive at the point (0, 0, 0)(0, 0, 0).

○
(0, 0, 0)(0, 0, 0) is a local maximum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(0, 0, 0)(0, 0, 0) is a saddle point.

1 point
Choose the correct statements about $f(x, y, z) = e^x(x^2 - y^2 - 2z^2)$f(x, y, z) = ex(x2 − y2 − 2z2)?

☐

(0, 0, 0)(0, 0, 0) is a local maximum.

(0, 0, 0)(0, 0, 0) is a saddle point.

Hessian test is inconclusive at the point (− 2, 0, 0)(−2, 0, 0).

(− 2, 0, 0)(−2, 0, 0) is a local maximum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(0, 0, 0)(0, 0, 0) is a saddle point.

(− 2, 0, 0)(−2, 0, 0) is a local maximum.

Differentiability for scalar valued multivariable function:

Differentiability for scalar valued multivariable function:

Let $f(x_1, x_2, \ldots, x_n)$f(x1, x2, … , xn) be a scalar valued function defined on a domain DD in R^n Rn containing an open ball around a point $a\sim = (a_1, a_2, \ldots, a_n)$∼a = (a1, a2, … , an). Then ff is differentiable at aa if

$$\lim_{h\sim \to 0} \frac{f(a\sim + h\sim) - f(a\sim) - h\sim.\nabla f(a\sim)}{||h\sim||} = 0$$

∼h→0lim ||∼h

||f(∼a + ∼h) − f(∼a) − ∼h.∇f(∼a) = 0

1 point

Which of the following functions are differentiable at (0, 0)(0, 0)?

$f(x, y) = x^2 + y^2$ f(x, y) = x2 + y2

$f(x, y) = (x\,y)^{\frac{1}{3}}$ f(x, y) = (xy)31

$f(x, y)$f(x, y) = {$\frac{x|y|}{\sqrt{x^2 + y^2}} if f(x, y) \neq (0, 0)$ $0 \quad if f(x, y) = (0, 0)$ { x2+y2

√

x|y|0if(x, y) = (0, 0)if(x, y) = (0, 0)

$f(x, y) = e^{x^2 + y^2}$ f(x, y) = ex2+y2

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y) = x^2 + y^2$ f(x, y) = x2 + y2

$f(x, y) = e^{x^2 + y^2}$ f(x, y) = ex2+y2

Check Answers

Your score is: 0/12